

Ang Cao

Research Scientist, Google DeepMind

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Research Appointments

Research Scientist, Google DeepMind

Dec. 2025–Present

- Self-improving AI systems for automating pretraining research.
- Spatial intelligence research for Gemini.
- Gemini 2D/3D agentic de-rendering via reinforcement learning for code generation with visual feedback.
- Gemini-Omni 3D/4D world modeling.

Research Scientist Intern, Meta FAIR, Menlo Park

May 2024–Dec. 2024

Mentors Sasha Sax and Franziska Meier.

Focus Proposed a 3D vision-language grounding pipeline using render-supervised learning from 2D foundation models, demonstrating strong data-scaling properties.

Research Scientist Intern, Meta GenAI, London

May 2023–Nov. 2023

Mentors David Novotny, Andrea Vedaldi, and Natalia Neverova.

Focus Designed a highly scalable GPU operator for neural-field rendering with over 1,000× lower memory usage than existing implementations, and developed feed-forward 3D reconstruction and generation methods.

Education

Ph.D., Computer Science and Engineering, University of Michigan

2020–2025 | GPA 4.0/4.0

Thesis: *Toward Spatial Intelligence via Data and Compute Efficiency*. Advisor: Justin Johnson.

Committee: Andrew Owens, Jeong Joon Park, Andrea Vedaldi, and Stella Yu.

M.S., Electrical and Computer Engineering, University of Michigan

2018–2020 | GPA 4.0/4.0

B.S., Electrical Engineering, Wuhan University

2014–2018 | GPA 3.9/4.0

Publications

* Equal contribution. † Mentored project. Citation record on Google Scholar.

[1] Xuanchen Lu, **Ang Cao**[†], Chao Feng, Andrew Owens.

“Generative Point Tracking and Forecasting.”

CVPR 2026 | Project

[2] Zichen Wang, **Ang Cao**[†], Liam J. Wang, Jeong Joon Park.

“MoE3D: A Mixture-of-Experts Module for 3D Reconstruction.”

arXiv 2026

[3] **Ang Cao**^{*}, Tiange Luo^{*}, Gunhee Lee, Justin Johnson, Honglak Lee.

“Probing Visual Language Priors in VLMs.”

ICML 2025 | Project

[4] **Ang Cao**, Sergio Arnaud, Oleksandr Maksymets, Jianing Yang, Ayush Jain, *et al.*.

“From Thousands to Billions: 3D Visual Language Grounding via Render-Supervised Distillation from 2D VLMs.”

ICML 2025 | Project

[5] Paul McVay, Sergio Arnaud, Ada Martin, Arjun Majumdar, . . . , **Ang Cao**, . . . , Franziska Meier.

“LOCATE 3D: Real-World Object Localization via Self-Supervised Learning in 3D.”

ICML 2025 | Project

[6] Jianing Yang, Alexander Sax, Kevin J. Liang, Mikael Henaff, Hao Tang, **Ang Cao**, Joyce Chai, Franziska Meier, Matt Feiszli.

“Fast3R: Towards 3D Reconstruction of 1000+ Images in One Forward Pass.”

CVPR 2025 | Project Code

Publications (continued)

- [7] **Ang Cao**, Justin Johnson, Andrea Vedaldi, David Novotny.
“Lightplane: Highly-Scalable Components for Neural 3D Fields.”
3DV 2025 | Project Code
- [8] Raphael Bensadoun, Tom Monnier, Yanir Kleiman, . . . , **Ang Cao**, . . . , Andrea Vedaldi.
“Meta 3D Gen.”
Technical report, 2024 | Project
- [9] Jiawei Ren^{*}, Liang Pan^{*}, Jiaxiang Tang, Chi Zhang, **Ang Cao**, Gang Zeng, Ziwei Liu.
“DreamGaussian4D: Generative 4D Gaussian Splatting.”
arXiv 2023 | Project
- [10] **Ang Cao**^{*}, Lukas Höllein^{*}, Andrew Owens, Justin Johnson, Matthias Nießner.
“Text2Room: Extracting Textured 3D Meshes from 2D Text-to-Image Models.”
ICCV 2023 Oral | Project Code
- [11] **Ang Cao**, Justin Johnson.
“HexPlane: A Fast Representation for Dynamic Scenes.”
CVPR 2023 | Project Code
- [12] **Ang Cao**, Chris Rockwell, Justin Johnson.
“FWD: Real-Time Novel View Synthesis with Forward Warping and Depth.”
CVPR 2022 | Project Code
- [13] **Ang Cao**, Justin Johnson.
“Inverting and Understanding Object Detectors.”
Technical report, 2021 | Paper Code
- [14] **Ang Cao**^{*}, Bowen Liu^{*}, Hun-Seok Kim.
“Unified Signal Compression Using Generative Adversarial Networks.”
ICASSP 2020 Oral | Paper

Honors and Awards

Outstanding Reviewer, IEEE/CVF Conference on Computer Vision and Pattern Recognition	2026
Rackham Conference Travel Grant, University of Michigan	2023
Rollin M. Gerstacker Foundation Fellowship, University of Michigan	2020
Outstanding Graduate of Wuhan University	2018
China National Scholarship, awarded to top 2% of undergraduates nationally	2016, 2017
Meritorious Winner, Mathematical Contest in Modeling	2017
National Undergraduate Innovation Foundation, Ministry of Education of China	2016

Professional Service

Conference reviewer: CVPR (since 2022), NeurIPS (since 2023), ICLR (since 2023), ECCV (since 2022), ICCV (since 2023), SIGGRAPH, and SIGGRAPH Asia.

Journal reviewer: IEEE TPAMI and IEEE TVCG.

Teaching

Head GSI / Teaching Assistant, EECS 498/598: Computer Graphics and Generative Models, University of Michigan Fall 2024

Invited Talks

“**Toward Building a Realistic Digital 3D World.**” NVIDIA Spatial Intelligence Lab; Meta FAIR; Google DeepMind; Meta Reality Labs; Stanford Vision and Learning Lab 2025